

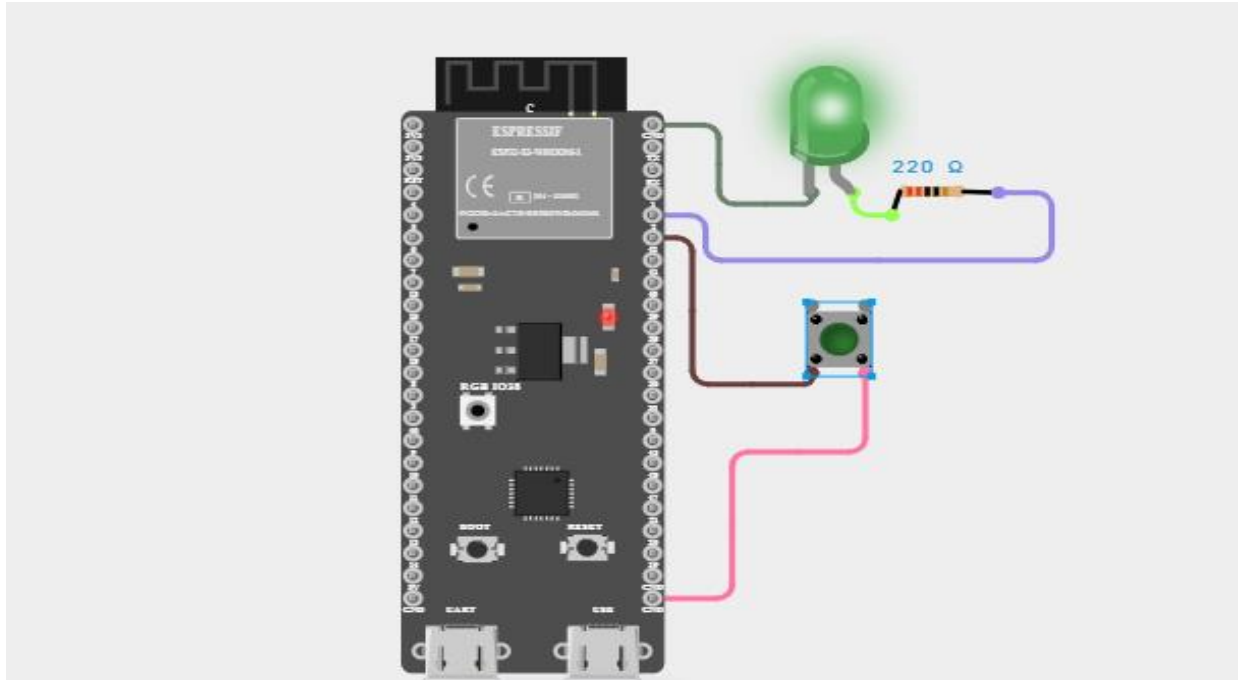
Embedded Computing Principles

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EXPERIMENT

LED Blinking & GPIO Control



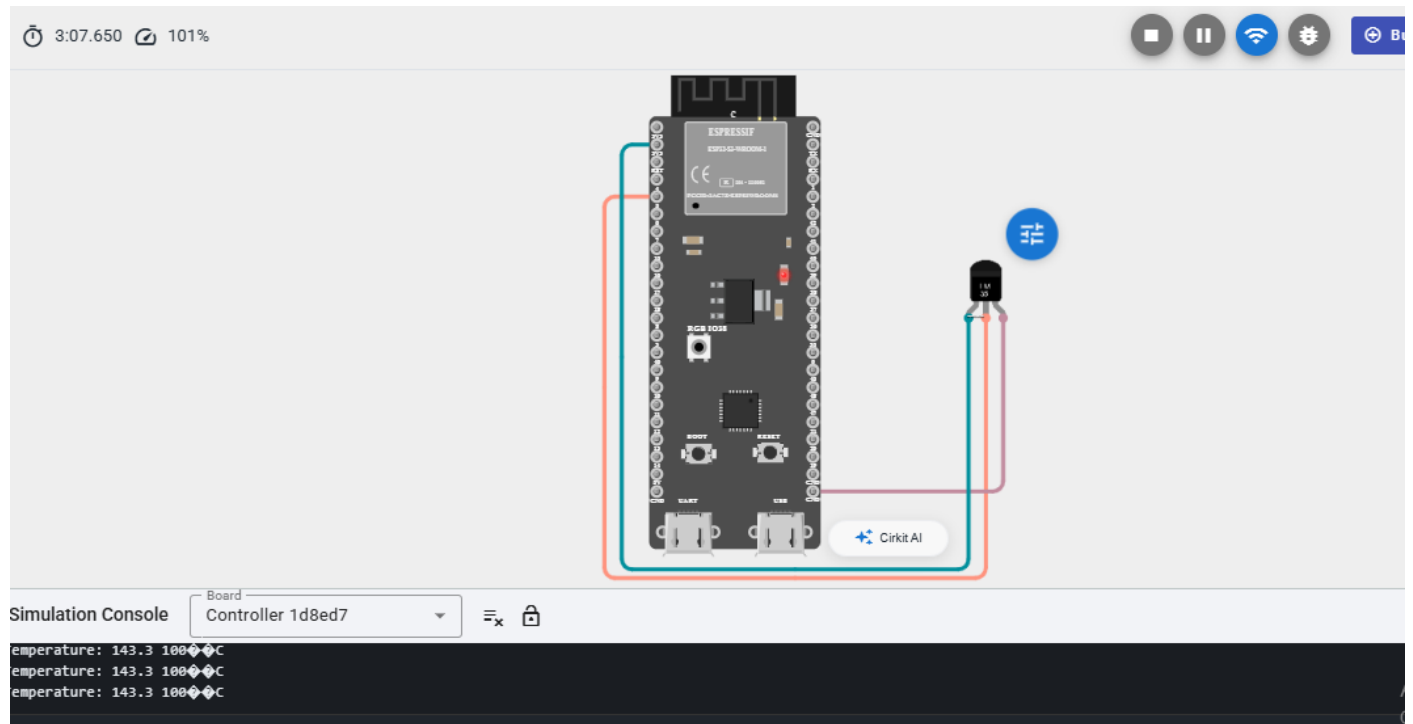
CODE:

```
const int LED_PIN = 2;
const int BUTTON_PIN = 42;

void setup() {
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUTTON_PIN, INPUT_PULLUP);
  digitalWrite(LED_PIN, LOW);
}

void loop() {
  if (digitalRead(BUTTON_PIN) == LOW) {
    digitalWrite(LED_PIN, HIGH);
    delay(250);
    digitalWrite(LED_PIN, LOW);
    delay(250);
  } else {
    digitalWrite(LED_PIN, LOW);
  }
}
```

Temperature Monitoring



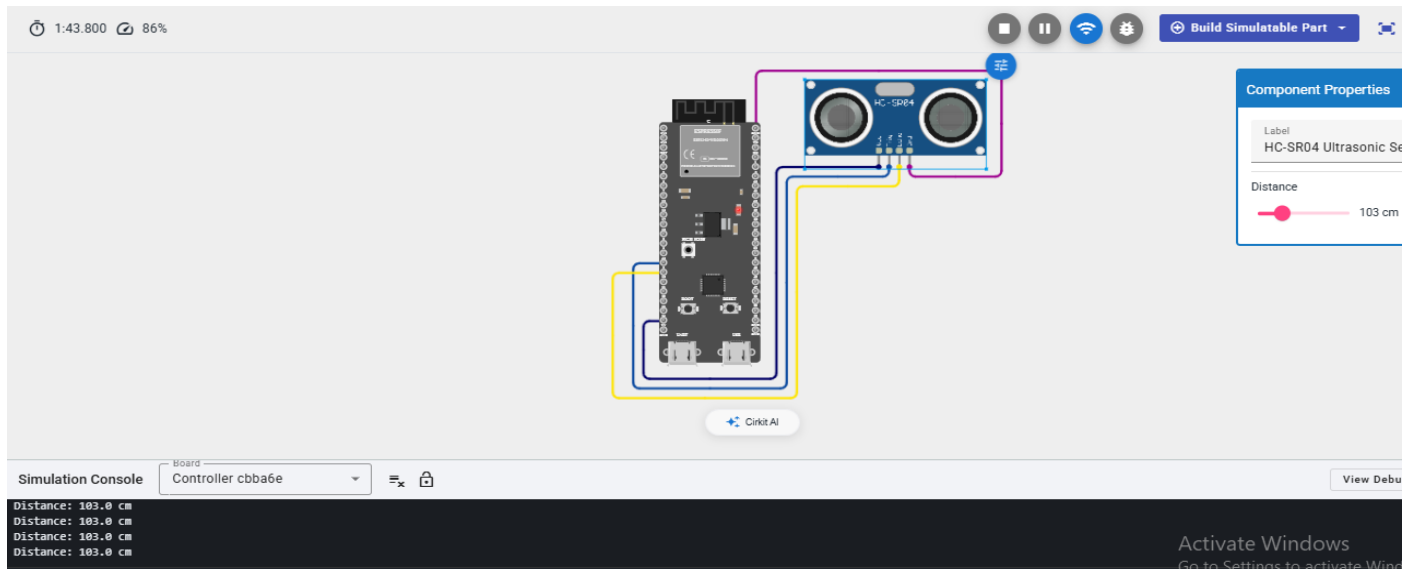
CODE:

```
#define LM35_PIN 4
```

```
void setup() {  
  Serial.begin(115200);  
  analogReadResolution(12);  
}
```

```
void loop() {  
  int adc = analogRead(LM35_PIN);  
  
  float voltage = adc * ( 3.3 / 4095.0);  
  float tempC = voltage * 100.0;  
  
  Serial.print("Temperature: ");  
  Serial.print(tempC, 1);  
  Serial.println(" 100°C");  
  
  delay(1000);  
}
```

Ultrasonic Distance Measurement

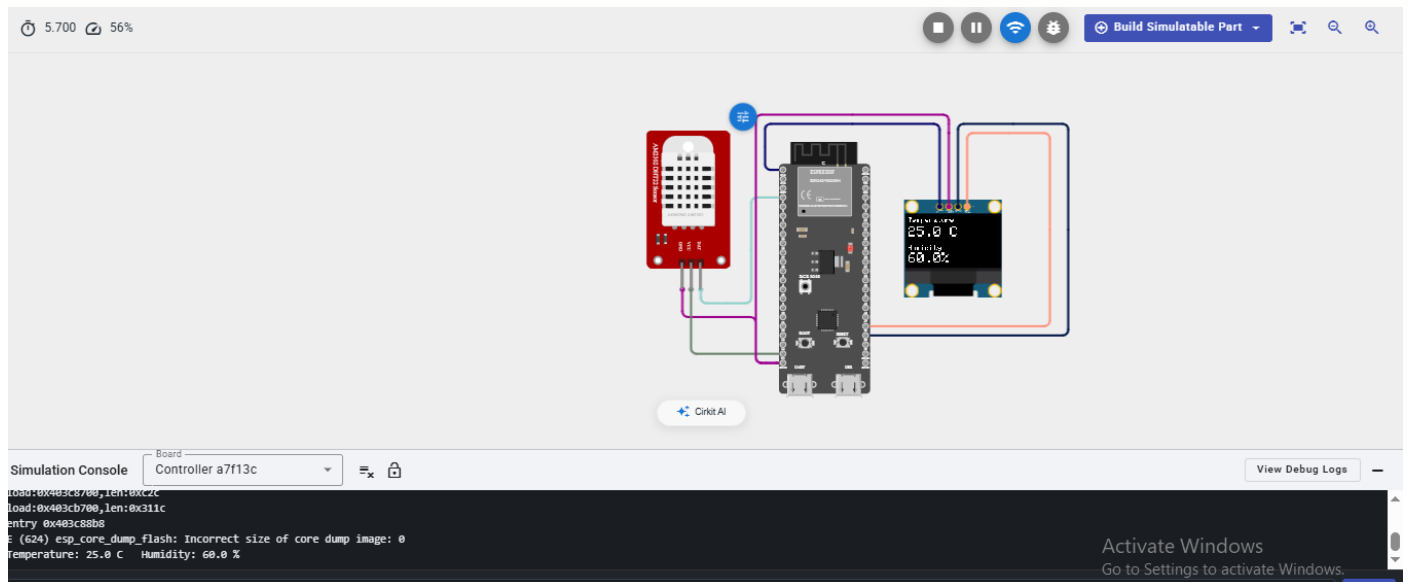


CODE:

```
#define TRIG_PIN 9
#define ECHO_PIN 10
void setup() {
  Serial.begin(115200);
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  digitalWrite(TRIG_PIN, LOW);
  Serial.println("ESP32-S3 Ultrasonic Distance Sensor");
}
void loop() {
  // Trigger pulse
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);
  // Measure echo
  long duration = pulseIn(ECHO_PIN, HIGH, 30000);
  if (duration == 0) {
    Serial.println("No Echo / Out of Range");
  } else {
    // Distance in cm
    float distance = duration * 0.0343 / 2.0;
    Serial.print("Distance: ");
    Serial.print(distance, 1);
    Serial.println(" cm");
  }

  delay(500);
}
```

OLED Display With with I2C



CODE:

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <DHT.h>

#define SDA_PIN 21
#define SCL_PIN 20

#define DHT_PIN 4
#define DHT_TYPE DHT22

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
#define OLED_ADDR 0x3C

DHT dht(DHT_PIN, DHT_TYPE);

Adafruit_SSD1306 display(
  SCREEN_WIDTH,
  SCREEN_HEIGHT,
  &Wire,
  OLED_RESET
);

void setup() {
  Serial.begin(115200);

  // OLED
  Wire.begin(SDA_PIN, SCL_PIN);
```

```
// DHT22
dht.begin();

// OLED initialization
if (!display.begin(SSD1306_SWITCHCAPVCC, OLED_ADDR)) {
  Serial.println("OLED not found!");
  while (1);
}

display.clearDisplay();
display.setTextColor(SSD1306_WHITE);

display.setTextSize(2);
display.setCursor(10, 20);
display.println("DHT22");
display.display();

delay(2000);
}

void loop() {

  float temperature = dht.readTemperature();
  float humidity = dht.readHumidity();

  // Check for invalid reading
  if (isnan(temperature) || isnan(humidity)) {
    Serial.println("DHT22 read failed");

    display.clearDisplay();
    display.setTextSize(2);
    display.setCursor(5, 20);
    display.println("DHT ERROR");
    display.display();

    delay(2500);
    return;
  }

  // Reject physically suspicious readings
  if (temperature < -40 || temperature > 80 ||
      humidity < 0 || humidity > 100) {

    Serial.println("Invalid DHT22 values");
```

```
display.clearDisplay();
display.setTextSize(2);
display.setCursor(5, 20);
display.println("BAD DATA");
display.display();

delay(2500);
return;
}

Serial.print("Temperature: ");
Serial.print(temperature, 1);
Serial.print(" C Humidity: ");
Serial.print(humidity, 1);
Serial.println(" %");

// OLED display
display.clearDisplay();

display.setTextSize(1);
display.setCursor(0, 0);
display.println("Temperature");

display.setTextSize(2);
display.setCursor(0, 12);
display.print(temperature, 1);
display.println(" C");

display.setTextSize(1);
display.setCursor(0, 38);
display.println("Humidity");

display.setTextSize(2);
display.setCursor(0, 48);
display.print(humidity, 1);
display.println("%");

display.display();

delay(2500);
}
```